Automatic Prompt Engineer

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Concept & Methodology Overview

- **Concept:** APE (Automatic Prompt Engineer) aims to find the optimal instruction (prompt) that leads to the best model performance.
- **Methodology:**
  - **Proposal:** Generate a set of candidate instructions using an LLM.
  - **Scoring:** Evaluate each candidate instruction based on a score function on the examples of tasks.
  - **Selection:** Choose the instruction with the highest score.

Mathematically: A "Black-Box" Optimization Problem

- APE converts instruction discovery into an optimization problem.
- We search for the instruction ρ that maximizes expected performance over input-output pairs (Q, A):

$$\rho^{\star} = \arg\max_{\rho} f(\rho) = \arg\max_{\rho} \mathbb{E}_{(Q,A)} \left[f_{\text{func}}(\rho, Q, A) \right]$$

APE Workflow: Step1

Step1: Generate Multiple Instruction candidates

- Given the Input/Output exemplars (Q, A)
- The LLM is prompt to create multiple instruction candidates ρ
- **Note:** Two approaches to generate high-quality candidates, one involved with a Masked Language Modeling task, the other involved with a text-continuation task

Forward Generation Template

I gave a friend an instruction and five inputs. The friend read the instruction and wrote an output for every one of the inputs. Here are the input-output pairs:

Input: $[Q_1]$ **Output**: $[A_1]$
Input: $[Q_2]$ **Output**: $[A_2]$
...

The instruction was **<COMPLETE>**

Reverse Generation Template

I instructed my friend to **<INSERT>**.

The friend read the instruction and wrote an output for every one of the inputs. Here are the input-output pairs:

Input: $[Q_1]$ **Output**: $[A_1]$
Input: $[Q_2]$ **Output**: $[A_2]$
...

APE Workflow: Step2

Step2: Rank Candidates Using Scoring Function

- For each instruction, prompt a target LLM with the instruction and inputs to generate outputs.
- Compare the generated outputs with the expected outputs to score each instruction.
  - Scoring Function: $\log P\left(A \mid [\rho; Q]\right)$
- The candidate instructions are ranked from highest scores (i.e. log probability) to lowest.

APE Workflow: Step3 (Optional)

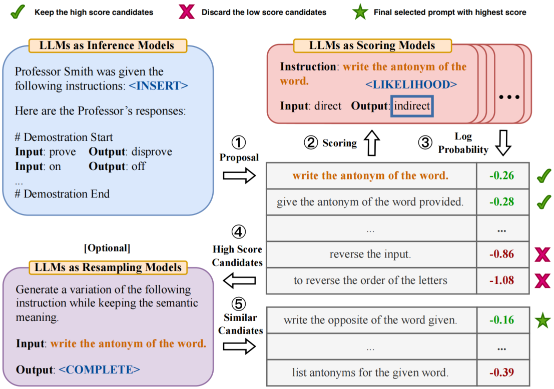
Step3: Re-sampling (Iterative Monte Carlo Search)

- Given the candidate Instruction selected by the previous step
- Use an LLM to generate new instructions similar to those with high scores completion

Note

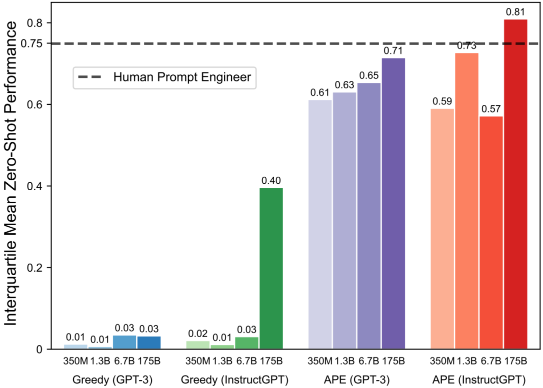
The author concluded that iterative generation provides **marginal improvement** over the relative simplicity and effectiveness of the generative process

APE Workflow Review: The whole story



- Each step is implemented as an instance of the pre-trained LLM's ability to complete text (or fill in a mask).
- No fine-tuning or adaptation of weights is involved.

APE Evaluation: Superhuman Performance!



(b) Interquartile mean across 24 tasks

- APE is able to surpass human performance when using the InstructGPT model.

Zero-shot: Improving on "Let's think step by step" – A demonstration of using an LLM to help craft better prompts to LLM's

Zero-shot: Ask a model to perform a task without any examples.

- Relies only on the model's internal knowledge and the clarity of the prompt.
- Standard zero-shot prompts often fail on tasks requiring reasoning, such as math, logic, or multi-step questions.

Solution: Use the phrase: "Let's think step by step"

After using APE to find a zero-shot prompt appendage that improves upon, APE creates:

- Let's work this out in a step by step way to be sure we have the right answer.
- The author's demonstrate improved performance on several benchmarks.

Summary

- **Novelty:**
  - Automatically discovers and ranks prompts without human-written examples or labeled data.
  - Evaluates prompts without training or updating model parameters.
  - Applies flexibly across tasks using only black-box language model access.
- **Comparison:**

Approach	Requires Examples	Manual Design	Adaptability
APE	No	No	Yes
Hard Prompting	No	Yes	No
Soft Prompting	No	Yes (learned)	Yes
In-Context Learning	Yes	Yes	Yes